Estimating Customer Lifetime Value using LRFM Model in Pharmaceutical and Medical Device Distribution Company

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***Abstract*— Dealing with intense competition encourages companies to build strong customer relationship management to improve customer satisfaction and retention. Extracting information related to customer lifetime value (CLV) supports the company to treat each customer differently based on their contribution to the profitability of the company. This study aims to estimate customer lifetime value of each cluster based on length, recency, frequency, and monetary (LRFM) model, hence provides the company an insight to evaluate their customers and improve marketing strategies. The proposed methodology is a combination of clustering k-means algorithm and AHP method to calculate the CLV using LRFM model in pharmaceutical and medical device distribution company. k-means algorithm was applied to cluster customers, while AHP method was used to gain information related to the importance of LRFM model. AHP results demonstrated that frequency is the most important variable in this sector. Finally, CLV was calculated and ranked for the resulting eight optimum clusters. Cluster 3, cluster 6, and cluster 8, which are the high value loyal customers, are the core customers of the company due to their CLV value. Therefore, the company should allocate company resources to focus on profitable customers.**

Keywords—clustering, customer lifetime value, customer segmentation, k-means, LRFM Model

I. Introduction

In recent years, companies are seeking innovative marketing strategies to improve customer satisfaction and retention due to intense competition. Pharmaceutical and medical device distribution company plays a vital role in fulfilling the needs of health care services. In Indonesia, both the pharmaceutical and medical devices industry are growing at warp speed as people recognize the lucrative part of this business. With the rapid growth in this industry, the competition among pharmaceutical and medical device distribution companies is getting more competitive. The high level of competition among pharmaceutical and medical device distribution companies forces the company to establish a long relationship with its customers. Extracting information regarding customer lifetime value (CLV) is essential in the sales activities of the company to generate personalized and targeted marketing strategies.

Customer Relationship Management (CRM) enables the company to use technology and human resources to develop insight related to the behavior of customers and the value of those customers [1]. CRM systems are divided into three general dimensions by namely: operational, analytical, and collaborative CRM [2]. Analytical CRM analyzes the characteristics of customer behavior by utilizing data mining tools to compare the characteristics of one customer with another, leading to the determination of customer segment and thus providing the basis for a targeted marketing campaign [3].

CLV is one of the popular approaches in CRM that is used as a tool to distinguish profitable customers, then rank the customers based on their contribution to the profitability of the company [4, 5]. CLV provides the company with the ability to treat each customer differently based on contribution rather than treating all the customers the same [5].

One powerful tool for assessing customer lifetime value is to perform customer segmentation using recency, frequency, and monetary (RFM) model [6]. CLV is based on the analysis of homogeneous groups instead of individual customers [7]. Customer segmentation allows companies to divide customers into different groups that have similar characteristics, thus enables the company to interact with each customer segment separately [8]. The RFM model was first introduced by Hughes (1994), in which customers are divided into five equal groups [9]. RFM model remains prominent in industry due to the ease of implementation in practice [10]. The LRFM model was then established by adding length to RFM model, which differs between the customer who has a long-term or short-term relationship with the company [11]. Table I shows the definition of LRFM attributes.

TABLE I. Definition of Attributes

Attributes	Definition
Length	Duration of the first and the last transaction in a certain period
Recency	Time interval since the last transaction to the observation period in this research (January 1, 2020) and gives information about buying potential of the customer
Frequency	The number of transactions within a certain period
Monetary	Total amount spent by customer during a certain period

Data mining technique, particularly the clustering technique, is widely employed in customer segmentation to divide all customers into the appropriate number of clusters based on some similarities between customers [12] [13]. There are several algorithms used in clustering. K-means algorithm is one of the popular approaches to perform cluster

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analysis due to its simplicity of implementation and fast execution [14].

This study aims to estimate the CLV of each cluster based on LRFM model using k-means clustering algorithm in the pharmaceutical and medical device distribution company. A previous study by [15] suggested that RFM variables are weighted differently depending on the characteristics of the industry. In this study, the importance of LRFM model was specifically assessed in the pharmaceutical and medical device distribution sector by using AHP method. In [9], CLV was evaluated using RFM model and fuzzy c-means clustering algorithm in IT company. A study by [16] estimated CLV using RFM model in health and beauty company. This study focusses on estimating CLV in a pharmaceutical and medical device distribution company that operates in Jakarta, Indonesia, based on LRFM model and k-means clustering algorithm.

The rest of this study is structured as follows: Section 2 gives an explanation on how the data are processed, Section 3 shows the results and analysis, and Section 4 presents the conclusion of this study.

II. Research Methods

Research methods consist of data preparation, clustering for customer segmentation, criteria weighting using AHP, and CLV calculation.

A. Data Preparation

Data preparation involves data collection, feature selection, and data transformation. The company investigated in this study runs in Business to Business (B2B) settings. The customers of the company are including pharmacy, government hospital, private hospital, clinic, laboratory, retailer, government institution, and private institution in Jakarta, Indonesia. The company acts as a distributor of pharmaceutical and medical device products that entrusted by manufacturers to sell their products to customers. Due to data availability, the proposed methods were applied to the transaction data set during the period between January 1, 2019, and December 31, 2019, which contains 7486 transactions from 589 customers.

The selected features, according to LRFM model, are the first transaction date during the period, the last transaction date, count of transactions, and total money that customer spend. Data normalization was then performed to change the data on the same scale [17]. Data normalization avoids significant difference in values between variables [17]. Data were normalized using min-max method as in (1).

$$\mathrm{v}' = \frac{\mathrm{v} - \min_{\mathrm{A}}}{\max_{\mathrm{A}} - \min_{\mathrm{A}}} (\mathrm{newmax}_{\mathrm{A}} - \mathrm{newmin}_{\mathrm{A}}) + \mathrm{newmin}_{\mathrm{A}} \quad (1)$$

Min-max normalization maps a value, v, of A to v' in the range of $[\mathrm{newmin}_{\mathrm{A}}, \mathrm{newmax}_{\mathrm{A}}]$ [16].

B. Clustering for Customer Segmentation

Clustering methods are classified into hierarchical methods and partitional methods. Partitional methods have the ability to analyze large data set because they do not require the calculation of similarity matrices among all observation, but instead just similarity of each observation to the cluster centroid [17]. This study applied one of the partitional methods, namely k-means algorithm to cluster the data. The quality of a cluster can be measured by within-cluster variation, which is a sum of squared error between all objects in cluster j and cluster centroid j, defined as the formula in (2).

$$J = \sum_{j=1}^{k} \sum_{i=1}^{n} [\![{x_i}^j - c_j]\!]^2 \quad (2)$$

From the equation above, J is the objective function of k-means clustering, ${x_i}^j$ is the i-th object in cluster j, c_j is the centroid of cluster j, k is the number of clusters, and n is the number of objects or members in the cluster j [18].

A common solution to find an optimum number of cluster *(k)* is by using cluster validation index which provides a score based on the concepts of compactness (i.e., how similar the members within the same cluster are) and separation (i.e., how dissimilar the members in different clusters are) [8]. This study used the Davies-Bouldin (DB) index to find an optimum number of clusters. The algorithm to calculate this index as in [19], can be described as follow. Define that C_i is the i-th cluster of the data and X_j indicates the j-th data sample with n attributes for the i-th cluster. Define S_i as in (3).

$$S_i = \left(\frac{1}{T_i} \sum_{j=1}^{T_i} \left[X_j - A_i\right]^q\right)^{\frac{1}{q}} \quad (3)$$

Where A_i is the center of i-th cluster and T_i shows the number of members inside that cluster [19]. $M_{i,j}$ is the distance between the two clusters i and j. $M_{i,j}$ calculation is based on the following formula:

$$M_{i,j} = \left(\sum_{k=1}^{n} \left[a_{k,i} - a_{k,j}\right]^p\right)^{\frac{1}{p}} \quad (4)$$

Where $a_{k,i}$ is the is the k-th attribute of the A_i element (center of the i-th cluster) [19]. $R_{i,j}$ is defined as an indicator of the quality or fairness of the clustering between the i-th and j-th clusters [19], and calculated using (5).

$$R_{i,j} = \frac{S_i + S_j}{M_{i,j}} \quad (5)$$

Davies-Bouldin index calculation with N total number of clusters is as in (6).

$$DB = \frac{1}{N} \sum_{i=1}^{N} D_i \quad (6)$$

Where D_i is the maximum value of $R_{i,j}$ $i \neq j$. The best choice of clusters, then, will be that which minimizes DB index value [20].

C. Criteria Weighting using AHP

AHP method determines the relative importance of variables based on the point of view of experts. Experts in sales and marketing divisions are asked to provide an assessment of the importance of the length, recency, frequency, and monetary criteria in Pharmaceutical and Medical Devices Distribution Companies. In using AHP method, the weights of importance of the variables are determined by using the pair-wise comparisons matrix.

D. CLV Calculation

A previous study by [16], used the weighted variables and normalized cluster centers for CLV calculation. CLV for each cluster is calculated using Equation as in (7).

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$$CLV_{ci} = NL_{ci}xWL_{ci} + NR_{ci}xWR_{ci} + NF_{ci}xWF_{ci} + NM_{ci}xWM_{ci} \quad (7)$$

Where NL_{ci} is the normal length, NR_{ci} is the normal recency, NF_{ci} is the normal frequency, and NM_{ci} is the normal monetary. Where WL_{ci} refers to the weighted length, WR_{ci} refers to the weighted recency, WF_{ci} refers to the weighted frequency, and WM_{ci} refers to the weighted monetary.

III. Results and Analysis

A. Selection Number of Clusters

This study executed k-means algorithm with a different number of clusters (k) ranging from two to ten. The maximum value of k was chosen as ten to divide the customers into a manageable number of groups. The cluster results were obtained for each k and were evaluated by using DB index as in Table II. Lower Davies-Bouldin index indicates better clustering quality [8]. As shown in Table II, DB index suggests eight optimum number of clusters.

TABLE II. Davies-Bouldin Index

k	Davies-Bouldin Index
2	0.965
3	1.094
4	1.055
5	0.933
6	0.920
7	0.913
8	0.887
9	0.939
10	0.966

B. Criteria Weights

The weighted LRFM variables were then calculated using the AHP method and represented as in Table III. The weighted LRFM results show that frequency is the most important variable (with criteria weight 0.431), followed by monetary, recency and length in this sector.

TABLE III. Criteria Weights

Variable	Criteria Weights
Length	0.114
Recency	0.146
Frequency	0.431
Monetary	0.308

C. CLV Results

CLV calculation used criteria weights and normalized cluster centers. Normalized cluster centers of each cluster, for CLV calculation were generated with LRFM variables, as shown in Table IV.

TABLE IV. Normalized Cluster Centers

Cluster	Length	Recency	Frequency	Monetary
1	0.869	0.903	0.235	0.077
2	0.667	0.647	0.134	0.078
3	0.924	0.957	0.779	0.722
4	0.038	0.378	0.012	0.082
5	0.132	0.867	0.050	0.059
6	0.928	0.916	0.614	0.160
7	0.454	0.203	0.089	0.059
8	0.829	0.850	0.268	0.588
Total	0.668	0.753	0.244	0.141

According to [11], the average LRFM value of each cluster, as shown in Table V, is compared with the total average LRFM to define clusters. If the average value of the cluster is greater than the total average, an upward arrow symbol (↑) is given. Meanwhile, the downward arrow symbol (↓) is given when the average value of the cluster is less than the total average. Cluster 1 refers to low consumption cost customer (LRFM, ↑↓↓↓). Customers in cluster 1 tend to have a close relationship with the company, but their consumption in the company is considered as low. The company needs to develop a marketing strategy to boost the spending of customers in cluster 1. Cluster 2, cluster 4, and cluster 7 belong to uncertain lost customer (LRFM, ↓↑↓↓). Cluster 2, cluster 4, and cluster 7 spend little money and rarely make purchases in the company. They have a small potential to become loyal according to the length and recency. The company should least focus on these clusters and better allocate its resources to more valuable clusters. Cluster 2 refers to the uncertain lost customer who gives a higher contribution to the profitability of the company compared to cluster 4 and cluster 7.

Referring to the CLV results in Table V, clusters whose higher CLV than the rests, are considered as the priority of the company because it shows a higher contribution to the profitability of the company. Cluster 3 has the highest CLV and gives the highest contribution to the profitability of the company, followed by cluster 6, cluster 8, cluster 1, cluster 2, cluster 5, cluster 7, and cluster 4.

Cluster 3, cluster 6, and cluster 8 belong to high value loyal customer (LRFM, ↑↓↑↑) and they tend to have a close and long relationship with the company. They often make purchases and spend much money in the company. Cluster 3, cluster 6, and cluster 8 are the core customers of the company due to their CLV value. Thus, the company should focus more on these clusters. Clusters 3 refers to the high value loyal customer who gives a higher contribution to the profitability of the company compared to cluster 6 and cluster 8. Cluster 5 belongs to uncertain new customer (LRFM, ↓↓↓↓). Customers in cluster 5 are the new customers in the company. They have not frequently interacted with the company yet, but the recency value indicates that cluster 5 has a high potential to repurchase in the company.

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TABLE V. CUSTOMER LIFETIME VALUE AND LRFM CHARACTERISTICS

Cluster	Customer Lifetime Value			LRFM Characteristics					
	CLV Results	CLV Rank	Total Customer	Average Length	Average Recency	Average Frequency	Average Monetary (IDR)	LRFM Symbol	Cluster Name
1	0.356	4	182	313.86	19.33	12.26	7,435,760	↑↓↓↓	Low Consumption Cost Customer
2	0.252	5	105	240.77	67.66	7.44	7,553,867	↓↑↓↓	Uncertain Lost Customer-Higher Contribution
3	0.804	1	15	333.66	9.06	38.4	69,317,609	↑↓↑↑	High Value Loyal Customer-Higher Contribution
4	0.090	8	47	13.63	118.59	1.55	7,891,885	↓↑↓↓	Uncertain Lost Customer-Lower Contribution
5	0.181	6	57	47.66	26.14	3.38	5,714,199	↓↓↓↓	Uncertain New Customer
6	0.554	2	91	334.86	16.80	30.49	15,380,486	↑↓↑↑	High Value Loyal Customer-Medium Contribution
7	0.138	7	49	163.83	151.57	5.26	5,675,444	↓↑↓↓	Uncertain Lost Customer-Medium Contribution
8	0.515	3	43	299.11	29.39	13.86	56,421,562	↑↓↑↑	High Value Loyal Customer-Lower Contribution
Average				241.30	47.61	12.70	13,559,770		

IV. CONCLUSION

CLV is widely applied to rank the customers based on their contribution to the profitability of the company. In this study, K-means clustering algorithm was applied to group customers based on their LRFM similarities, and there were eight resulting clusters obtained. AHP method weighted the LRFM model, and the results show that frequency is the most important variable in the pharmaceutical and medical device distribution sector, followed by monetary, recency, and length. CLV ranking shows that cluster 3 has the highest contribution to the profitability of the company, followed by cluster 6, cluster 8, cluster 1, cluster 2, cluster 5, cluster 7, and cluster 4. Customers in cluster 3, cluster 6, and cluster 8 are the core customers of the company because they give higher profitability to the company. The company should focus and allocate its resources more on these profitable clusters.

This study has some limitations, such as the weighted LRFM model, which was carried out specifically in pharmaceutical and medical device distribution company. Future research should develop the weighted LRFM model according to the selected sector.

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